FRANK TAMBURRO

ECE 121 (Spring 2025)
Lab#1 (Introductory Labs)

Please answer the following problems and write a report. You can work in a group (max 2 people) or you can work individually

Report writing format: (Objectives, Introduction, Results& Analysis (answer the following problems), Conclusion & Discussion)

Submission Format: One pdf file.

<u>Problem#1 (Introduction to experimental components/instruments): (10 points)</u>
The goal of this part is to introduce the various components and instruments that will be used in the ECE 121 labs for constructing, powering, and characterizing the various circuits. These include:

Breadboard for circuit construction/assembly
▪ DVM or DMM (Digital Multi-Meter) measuring instrument
▪ DC Sources: Voltage Source

>Action Item 1.1: In your report, in the introduction part, write a one-paragraph description and at least one photograph for each of the items introduced in this part of the lab.

In this lab we are working with making a simple circuit that contains 1 DC power source, 2 resistors, and a ground. In the lab and on Pspice we find the current and voltage across each of the resistors using DMM's. The overall point of this lab is to get comfortable with the measuring tools, breadboards, and how to use Pspice. All of the photos for the problems are at the bottom of the document.

<u>Problem#2 (Introduction to DC measurements and testing of resistive circuits at DC conditions): (30 points)</u>

The goal of this part is to learn how to use a DVM/DMM to measure the following:
>Action Item: 2.1 Pick 2 resistances and measure the value of resistance.

R1= <u>4.801Ω?</u>

R2= <u>2.178KΩ?</u>

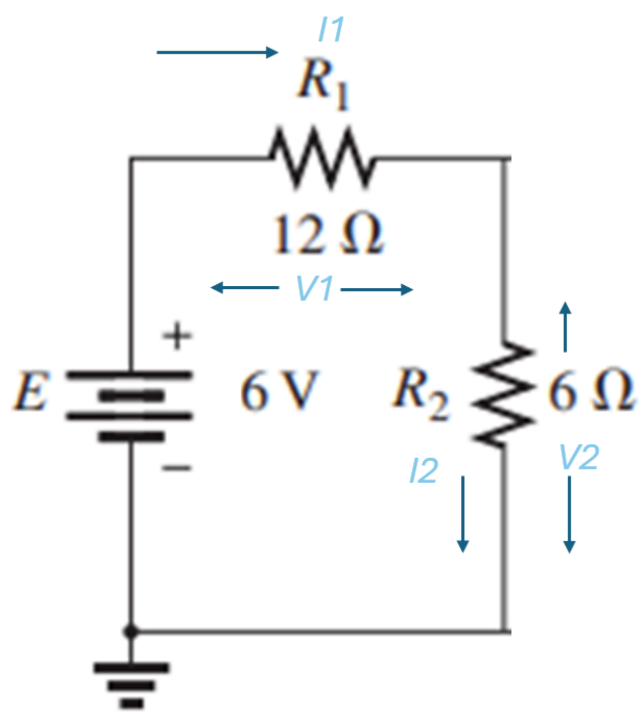
>Action Item: 2.2 Measure the value of an equivalent resistance: Resistors in series (10 points)
Combine the two resistors in series

on a breadboard and measure the equivalent resistance. Req-s= ____2.182KΩ?

Problem#3 (Build the circuit in the Lab and measure voltage and current): (30 points)

- Build the following circuit in the lab to perform an appropriate simulation to determine current I1, I2, voltage V1, and V2 across that branch. You take two resistors from the lab and measure the resistance. You use that resistance to build the circuit.

>Action Item: 3.1 Please add your circuit-built image in your report which shows the circuit depicting voltage, current, and resistors. Add your hand calculation screenshot with the report.



>Action Item: 3.2 Measure the value of a voltage and current

I1= __0.001mA______?

I2= __0.001mA________?
V1= _12.303mV_____?

V2= _5.614V________?

Problem#4 (Build the circuit in the PSpice and measure voltage and current): (40 points)

- Draw the following circuit in the PSpice/Circuit simulator to perform an appropriate simulation to determine current I1, I2, voltage V1, V1 across that branch. Use the same resistor value that you used for problem 1.

>Action Item: 4.1 Please add your circuit simulated screenshot in your report which shows the circuit depicting voltage and current.

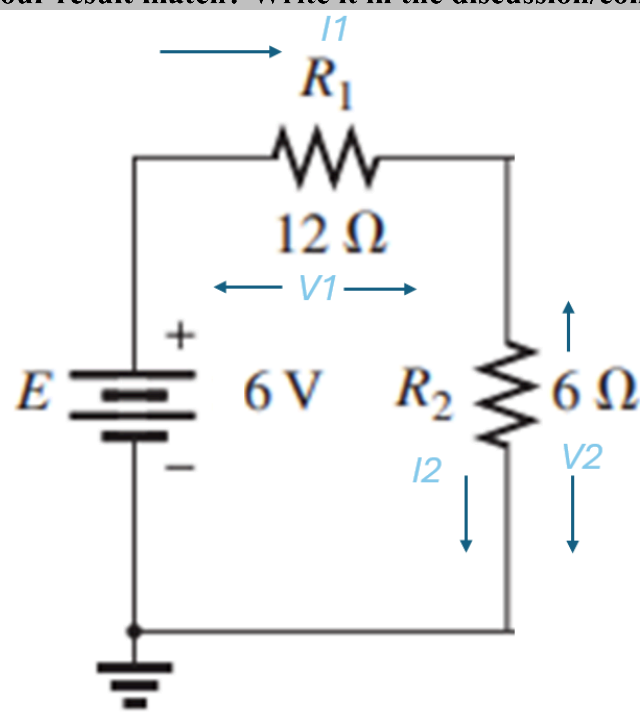
>**Action Item: 4.2 Measure the value of a voltage and current**

I1= 2.751mA ?

I2= 2.751mA ?
V1= 0.013 ?

V2= 5.987V ?

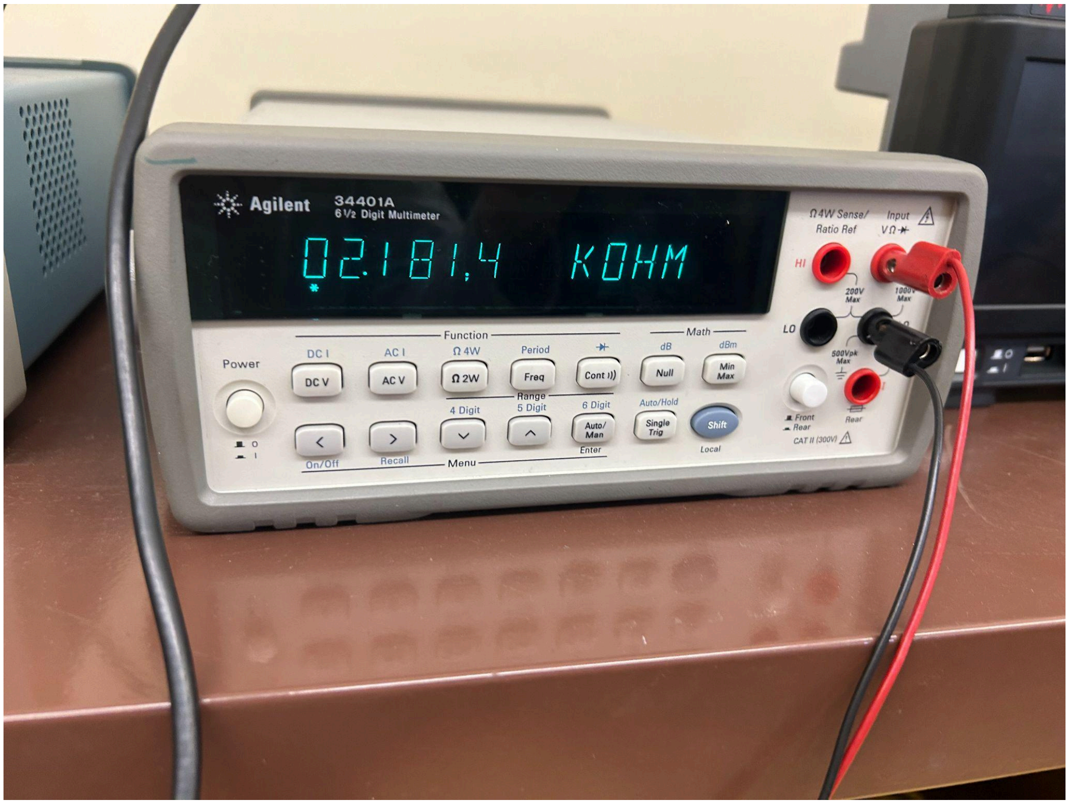
>Action Item: 4.3 Compare your calculated, measured, and simulated results in the following table. Does your result match? Write it in the discussion/conclusion section.

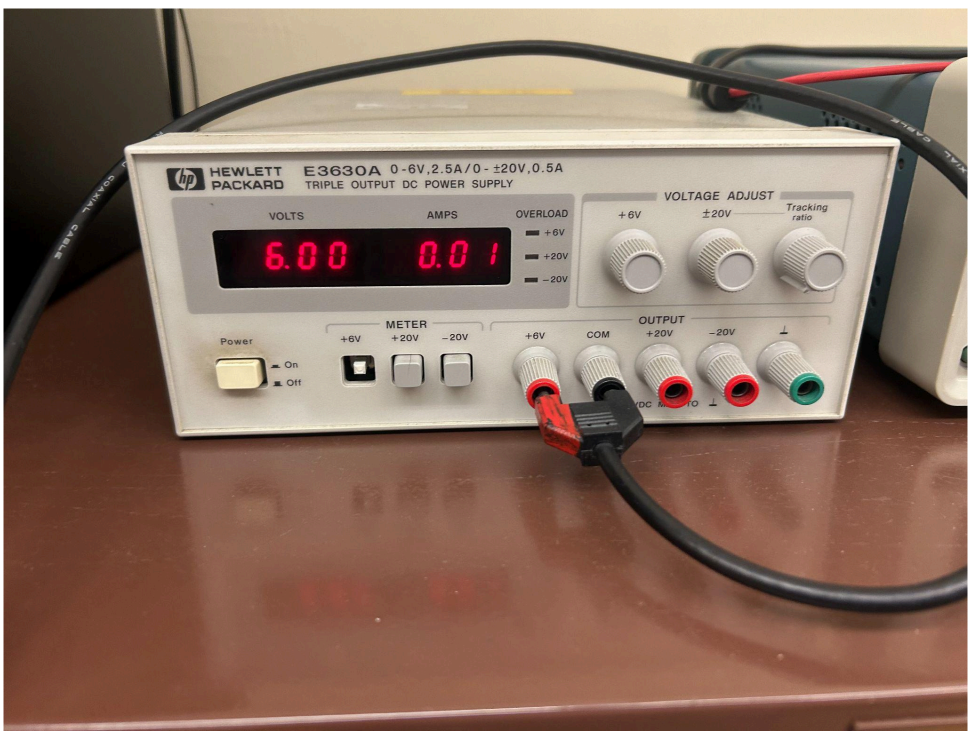


	Calculated	Measured	Simulated
R1	4.801Ω	4.801Ω	No
R2	2.178KΩ	2.178KΩ	No
Req	2.182KΩ	2.182KΩ	No
I1	2.749mA	0.001mA	2.751mA
I2	2.749mA	0.001mA	2.751mA
V1	13.201mV	12.303mV	13.0mV
V2	5.989V	5.614V	5.987V

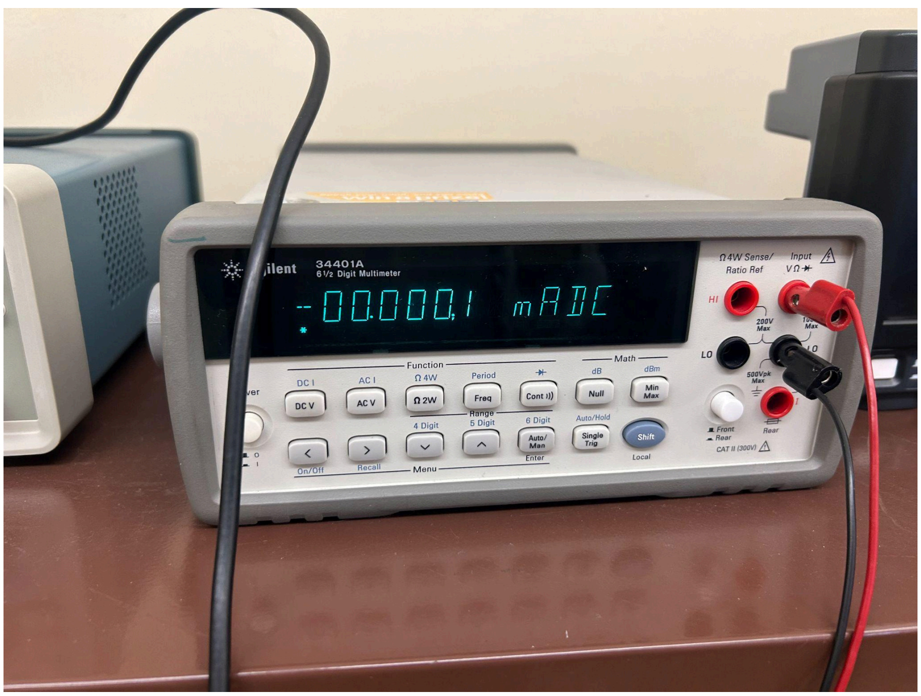

Agilent
34401A
6 1/2 Digit Multimeter
021765 KOHM
Function
Power
DC V
AC V
Ω 2W
Freq
Null
Shift


Agilent
34401A
6 1/2 Digit Multimeter
004.802 OHM
Function
Math
Power
DC I
AC I
Ω 4W
Period
dB
dBm
DC V
AC V
Ω 2W
Freq
Cont
Null
Min Max
Range
4 Digit
5 Digit
6 Digit
Auto/Hold
Auto/ Man
Single Trig
Shift
Enter
Local
On/Off
Recall
Menu
HI
LO
Ω 4W Sense/ Ratio Ref
Input
200V Max
1000V Max
500Vpk Max
Front
Rear
CAT II (300V)


Agilent
34401A
6½ Digit Multimeter
02.181,4 KOHM
Function
Math
DC I
AC I
Ω 4W
Period
dB
dBm
Power
DC V
AC V
Ω 2W
Freq
Cont
Null
Min Max
Range
4 Digit
5 Digit
6 Digit
Auto/Hold
Single Trig
Shift
Auto/ Man
Enter
On/Off
Recall
Menu
Local
Ω 4W Sense/ Ratio Ref
Input VΩ
HI
LO
200V Max
1000V Max
500Vpk Max
Front
Rear
CAT II (300V)


HEWLETT PACKARD
E3630A 0-6V,2.5A/0-±20V,0.5A
TRIPLE OUTPUT DC POWER SUPPLY
VOLTS
AMPS
OVERLOAD
+6V
+20V
−20V
6.00
0.01
VOLTAGE ADJUST
+6V
±20V
Tracking ratio
Power
On
Off
METER
+6V
+20V
−20V
OUTPUT
+6V
COM
+20V
−20V


Agilent
34401A
6½ Digit Multimeter
012.303 mVDC


34401A
6½ Digit Multimeter
-00.0001 mADC


Agilent
34401A
6½ Digit Multimeter
004.802 OHM

Frank Tamburro

$$R_{eq} = 2.178\,k\Omega + 4.801\,\Omega$$
$$= 2.182\,k\Omega$$

$$I_1 = \frac{V}{R} \qquad V = IR$$

$$I_1 = \frac{6V}{2.182\,k\Omega} = 0.002749A$$

$$I_2 = 0.002749A$$

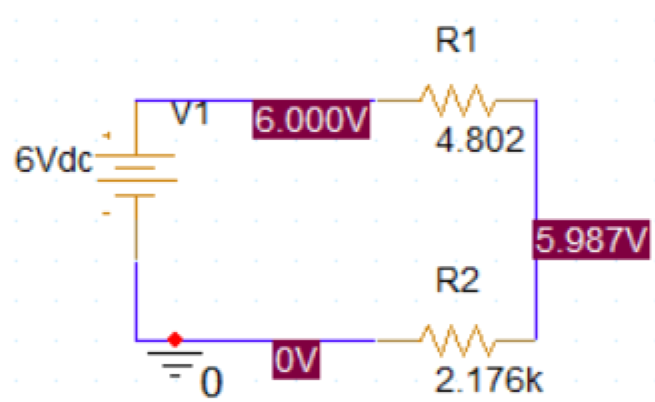
$$V_1 = V_s \frac{4.801}{4.801 + 2.178k}$$

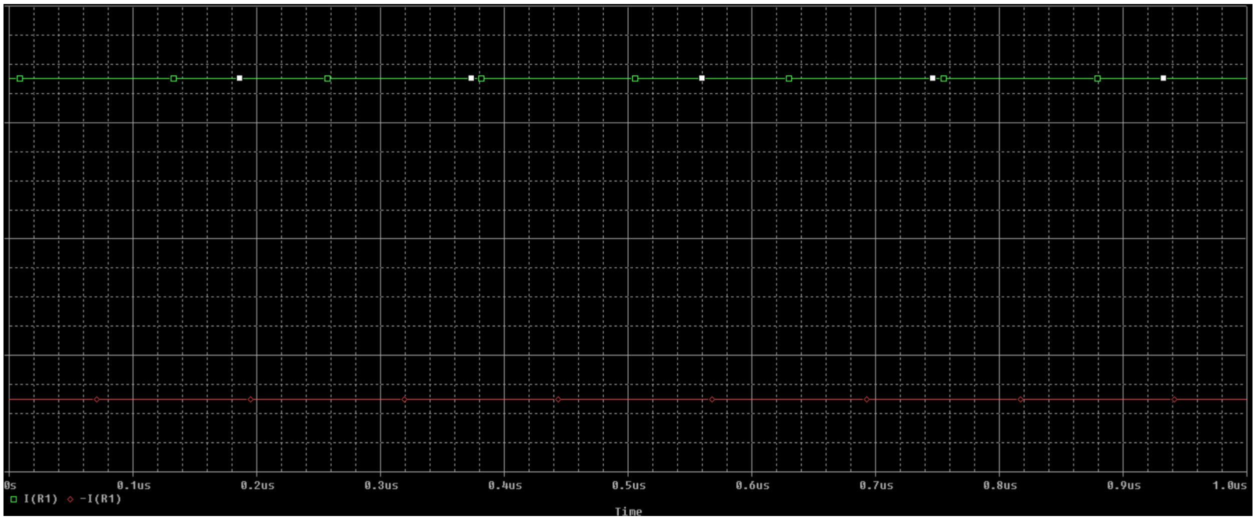
$$V_1 = 6 \frac{4.801}{4.801 + 2.178\,k\Omega}$$

$$V_1 = 0.013201V$$

$$V_2 = V_s \frac{2.178k}{4.801 + 2.178k}$$

$$V_2 = 6 \frac{2.178k}{2.182k} = 5.989V$$





	A
	SCHEMATIC1 : N00099
BiasValue Current	2.751mA
BiasValue Voltage	6.000V
Color	Default
Line Style	Default
Line Width	Default
Name	N00099